MIL-I-36013B
13 July 1972
SUPERSEDES
MIL-I-36013A
2 November 1965

MILITARY SPECIFICATION

ILLUMINATOR, X-RAY FILM, COLLAPSIBLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a lightweight, collapsible, X-ray film illuminator for use in the field for viewing films up to and including 14 inches by 17 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-529	Enamel Alkyd, Semi-Gloss.
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Fiber.

FSC 6525

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

Military

MIL-C-5541	Chemical Films and Chemical Film materials for Aluminum and Aluminum Alloys.
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-P-8585	Primer Coating Zinc Chromate, Low-Moisture-Sensitivity.

STANDARDS

Federal

Fed-Std-141	Paint, Varnish, Laquer, and Related Materials; Methods of Inspection, Sampling and Testing.
Fed-Std-595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

20058	Illuminator, X-Ray Film, Incandescent, Collapsible, Field, 110 Volt, AC-DC.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fire and casualty hazards. The contractor shall submit proof prior to delivery that all electrical components used in the construction of the illuminators conform to the requirements of the Underwriter's Laboratories, Inc. (UL). UL listing of the component shall be acceptable as proof of compliance. In lieu of UL listing a laboratory report from a qualified independent laboratory, adequately equipped and competent to determine compliance with UL standards, shall be acceptable. Any laboratory report submitted shall include actual test methods, data taken, and results of tests and examinations. The laboratory report shall show compliance of the component with the applicable UL standard in order to be acceptable.

3.2 Material. Material shall be of the highest quality throughout. Drawing 20058, constitutes the basic requirements for the fabrication of the illuminator.

3.3 Design and construction. The illuminator shall be in strict accordance with Drawing 20058 and shall accommodate radiographic film up to and including 14 inches by 17 inches. The illuminator shall collapse and reopen without affecting the fit of the component parts. The unit shall operate satisfactorily on 110 to 125 volts AC-DC, and shall utilize two medium base 115 volt, 100 watt incandescent lamps as a light source. The viewing screen shall be a single translucent, opal, methyl methacrylate polymer diffusing view panel with a light transmission factor of 55 $\pm$ 5 percent. The ground wire of the power cord shall be securely fastened to the frame of the illuminator.

3.4 Dielectric strength. The illuminator shall exhibit no electrical breakdown when tested as in 4.4.1.

3.5 Finish. Prior to installation of the electrical components, the illuminator shall be thoroughly cleaned and chemically degreased. After degreasing, the surfaces shall be coated in accordance with MIL-C-5541. This coating shall have a clear color.

3.5.1 Exterior. The exterior of the illuminator shall be finished with acid, alkali, and solvent-resisting enamel, as follows:

A zinc-chromate primer conforming to MIL-P-8585 shall be applied first The primer coat shall form a grounding bond and base for the finishing enamel. The primer coat shall be followed by two coats of semi-gloss enamel conforming to class B of TT-E-529. Each coat shall be sprayed on and baked in high temperature ovens in accordance with the paint manufacturer's recommendations. The resulant finish shall be hard, smooth, resistant to abrasion, chipping, and cracking. The color of the enamel shall be green, standard color chip 24087, as shown in FED-STD-595 (see 6.6). A final coat of weather resisting, clear, synthetic baking enamel shall be applied and baked in accordance with the paint manufacturers recommendations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

3 5.2 Interior The interior of the illuminator shall not be treated further than the coating in accordance with MIL-C-5541.

3.6 Identification marking. Each illuminator shall have a nameplate, permanently and legibly marked with at least the following information:

Federal Stock Number
Manufacturer's name or registered trademark
Contract Number
110 Volt AC-DC
U.S.

The use of adhesives, to attach the nameplate, shall not be acceptable. As an alternate to the use of a nameplate, the identification information shall be impression stamped on the back of the illuminator. Nameplate shall be a minimum of .020 inch thick.

3.7 Service data The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification Identical or interchangeable replacement parts shall be identified by a single part number only As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, two preliminary copies of his proposed service data, for review and comment The preliminary service data shall be submitted sufficiently early to permit adequate review (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

3.8 Workmanship. Workmanship shall be first class throughout. Illuminators shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time or from time to time during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the aluminum, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the Federal Stock Number (FSN), the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

4.2 Sampling The unit of product for inspection purposes shall be one illuminator.

* 4.2.1 For examination Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I

Table I Sampling for Examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-2	2.5

* 4.2.2 For tests All illuminators (100 percent) supplied shall be tested as specified herein

4.3 Examination The illuminators shall be examined to determine compliance with all requirements contained in this specification

* 4.3.1 Classification of defects Examination shall be conducted in accordance with the following classification of defects as shown in Table III

Table III Categories and defects*

Categories	Defects
Major	
101	Illuminator not complete with all specified component parts
102	Film clips do not retain, in their proper position, standard size films up to 14 inches by 17 inches
103	Exposed or accessible edges and corners of component parts not free of burs, sharpness
104	Any metal part not free of cracks, tears, ruptures, ripples, noticeably deformed areas
105	Spot welds not present as required, not secure
106	Component parts not properly fitted and assembled into unit
107	Any hinge not operating smoothly, not adequately secured against removal
108	Illuminator not easily collapsed, not easily opened.
109	Viewing screen not free of cracks ruptures, ripples.
Minor	
201	Finish not free of scratches, marred area, thin areas.
202	Color not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

Table III. Categories and defects*

Categories	Defects
Minor	
203	Identification marking not complete, legible, permanent.
204	Service data missing, not complete.
205	Marking on unit carton not present, legible, complete.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The illuminator shall be examined for compliance with dimensional requirements. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Dielectric strength. The illuminator shall be subjected to a gradually increasing potential of twice the rated voltage plus 1000 volts which shall be maintained at this value for a period of one minute. During this period, instrument used for test shall be checked continually for evidence of breakdown.

5 PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One illuminator, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	4 units

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table apply to PPP-B-636 box.

5.2 1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2 2 Level B

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type, PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be method II for PPP-B-636 box. Special requirements of applicable table apply to PPP-B-636 box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container Exterior container shall be marked in accordance with MIL-STD-129

6. NOTES

* 6.1 Intended use X-Ray illumination in field military medical facilities

6.2 Ordering data Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing (see 5.1 - 5.2)

6.3 This specification does not include all types classes, grades, and sizes, of the commodity indicated by the title of this specification or; which are commercially available, but is intended to cover one type which is normally procured to meet Military requirements

* 6.4 This specification covers the following item appearing in the Federal Supply Catalog

FSN	Item Identification
6525-209-7216	Illuminator, X-Ray Film, Incandescent Illuminated, Collapsible, 110 Volt, AC-DC

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

6.5 Color chips referenced in FED-STD-595 are available form the Naval Supply Depot, Philadelphia, Pa.

* 6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No 6525-1098

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors government activities or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

MIL-I-36013B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.

FOLD

DFFFNSF PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa 19101

Defense Supply Agency

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STRFET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:57Z
Check the source to verify that this is the current version before use.